

Exercise J01

Q1. Which one of the following is TRUE.

- (A) $x \cdot \int \frac{dx}{x} = x \ln|x| + C$ (B) $x \cdot \int \frac{dx}{x} = x \ln|x| + Cx$ (C) $\frac{1}{\cos x} \cdot \int \cos x \, dx = \tan x + C$
(D) $\frac{1}{\cos x} \cdot \int \cos x \, dx = x + C$

Ans. B

Sol. -

Q2. If $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + C$, then value of (A,B) is-

- (A) $(\sin \alpha, \cos \alpha)$ (B) $(\cos \alpha, \sin \alpha)$ (C) $(-\sin \alpha, \cos \alpha)$ (D) $(-\cos \alpha, \sin \alpha)$

Ans. B

Sol. -

Q3. $\int \frac{dx}{\cos x + \sqrt{3} \sin x} =$

- (A) $\frac{1}{2} \log \tan \left(\frac{x}{2} + \frac{\pi}{12} \right) + C$ (B) $\frac{1}{2} \log \tan \left(\frac{x}{2} - \frac{\pi}{12} \right) + C$ (C) $\log \tan \left(\frac{x}{2} + \frac{\pi}{12} \right) + C$
(D) $\log \tan \left(\frac{x}{2} - \frac{\pi}{12} \right) + C$

Ans. A

Sol. -

Q4. If $\int \frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1} = C - \frac{1}{f(x)}$ where $f(x)$ is of the form of $ax^2 + bx + c$ then $(a+b+c)$ equals

- (A) 4 (B) 5 (C) 6 (D) None

Ans. B

Sol. $\int \frac{2x+3}{(x^2+3x)(x^2+3x+2)+1} dx$

put $x^2+3x=t \Rightarrow (2x+3) dx = dt$

$$\int \frac{dt}{t(t+2)+1}; \int \frac{dt}{(t+1)^2} = C - \frac{1}{t+1} = C - \frac{1}{x^2+3x+1}$$

$$\Rightarrow a=1, b=3, c=1 \Rightarrow a+b+c=5$$

Q5. Consider $f(x) = \frac{x^2}{1+x^3}$; $g(t) = \int f(t) dt$. If $g(1) = 0$ then $g(x)$ equals

(A) $\frac{1}{3}\ln(1+x^3)$ (B) $\frac{1}{3}\ln\left(\frac{1+x^3}{2}\right)$ (C) $\frac{1}{2}\ln\left(\frac{1+x^3}{3}\right)$ (D) $\frac{1}{3}\ln\left(\frac{1+x^3}{3}\right)$

Ans. B

Sol. -

Q6. $\int \frac{\cot^{-1}(e^x)}{e^x} dx =$

(A) $\frac{1}{2}\ln(e^{2x}+1) - \frac{\cot^{-1}(e^x)}{e^x} + x + c$ (B) $\frac{1}{2}\ln(e^{2x}+1) + \frac{\cot^{-1}(e^x)}{e^x} + x + c$
 (C) $\frac{1}{2}\ln(e^{2x}+1) - \frac{\cot^{-1}(e^x)}{e^x} + x + c$ (D) $\frac{1}{2}\ln(e^{2x}+1) + \frac{\cot^{-1}(e^x)}{e^x} - x + c$

Ans. C

Sol.
$$\begin{aligned} \int \frac{\cot^{-1}(e^x)}{e^x} dx &= \int \underbrace{\cot^{-1}(e^x)}_I \cdot \underbrace{e^{-x}}_{II} dx \\ &= \cot^{-1}(e^x) \cdot (-1)e^{-x} + \int \frac{-1}{1+e^{2x}} e^{-x} \cdot e^x dx \\ &= -\cot^{-1}(e^x) \cdot e^x - \int \frac{dx}{1+(e^x)^2} \\ &= \frac{-\cot^{-1}(e^x)}{e^x} - \frac{1}{2} \int \frac{2dx}{(1+(e^x)^2)} \\ &= \frac{-\cot^{-1}(e^x)}{e^x} - \frac{1}{2} \int \frac{2e^{-2x}}{e^{-2x}+1} dx \\ &= \frac{-\cot^{-1}(e^x)}{e^x} + \frac{1}{2} \ln(e^{-2x}+1) + C \\ &= \frac{-\cot^{-1}(e^x)}{e^x} + \frac{1}{2} \ln(1+e^{2x}) - x + C \end{aligned}$$

Q7. $\int x \cdot \frac{\ln(x+\sqrt{1+x^2})}{\sqrt{1+x^2}} dx$ equals :

(A) $\sqrt{1+x^2}\ln(x+\sqrt{1+x^2}) - x + c$ (B) $\frac{x}{2} \cdot \ln^2(x+\sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + c$
 (C) $\frac{x}{2} \cdot \ln^2(x+\sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} + c$ (D) $\sqrt{1+x^2}\ln(x+\sqrt{1+x^2}) + x + c$

Ans. A

Sol. Let's group the terms to easily identify our parts for integration by parts:

$$\int \ln(x+\sqrt{1+x^2}) \cdot \left(\frac{x}{\sqrt{1+x^2}}\right) dx$$

Apply the Integration by Parts Formula

Substituting u , v , du , and dv into the formula:

$$\int x \cdot \frac{\ln(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx = \sqrt{1+x^2} \cdot \ln(x + \sqrt{1+x^2}) - \int \sqrt{1+x^2} \cdot \left(\frac{1}{\sqrt{1+x^2}}\right) dx$$

$$= \sqrt{1+x^2} \cdot \ln(x + \sqrt{1+x^2}) - \int 1 dx = \sqrt{1+x^2} \cdot \ln(x + \sqrt{1+x^2}) - x + C$$

Q8. $\int \frac{(\log x)}{x^2} dx =$

(A) $\frac{1}{2}(\log x + 1) + c$ (B) $-\frac{1}{x}(\log x + 1) + c$ (C) $-\frac{1}{x}(\log x - 1) + c$ (D) $\log(x + 1) + c$

Ans. B

Sol. -

Q9. If $\int f(x) dx = \Psi(x)$, then $\int x^5 f(x^3) dx$ is equal to :

(A) $\frac{1}{3} \left[x^3 \Psi(x^3) - \int x^2 \Psi(x^3) dx \right] + C$ (B) $\frac{1}{3} x^3 \Psi(x^3) - 3 \int x^3 \Psi(x^3) dx + C$

(C) $\frac{1}{3} x^3 \Psi(x^3) - \int x^3 \Psi(x^3) dx + C$ (D) $\frac{1}{3} \left[x^3 \Psi(x^3) - \int x^3 \Psi(x^3) dx \right] + C$

Ans. C

Sol. -

Q10. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx$

(A) $2e^{\sqrt{x}}[x - \sqrt{x} + 1] + C$ (B) $e^{\sqrt{x}}[x - 2\sqrt{x} + 1]$ (C) $e^{\sqrt{x}}(x + \sqrt{x}) + C$

(D) $e^{\sqrt{x}}(x + \sqrt{x} + 1) + C$

Ans. A

Sol. $I = \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx$

let $\sqrt{x} = t$

$\Rightarrow \frac{1}{2\sqrt{x}} dx = dt$

$I = 2 \int \underbrace{e^t}_{II} \underbrace{(t^2 + t)}_I dt$

$= 2 \left[e^t (t^2 + t) - \int (2t + 1) e^t dt \right]$

$= 2e^t (t^2 + t) - 2(2t + 1)e^t + 2 \int 2e^t dt$

$= 2[e^t (t^2 + t) - (2t + 1)e^t + 2e^t] + C$

$$= 2 \left[e^{\sqrt{x}} (x + \sqrt{x} - 2\sqrt{x} - 1 + 2) \right] + C$$

$$= 2e^{\sqrt{x}} (x + 1 - \sqrt{x}) + C$$

Q11. $\int \left\{ \frac{(\log x - 1)}{1 + (\log)^2} \right\}^2 dx$ is equal to -

(A) $\frac{\log x}{(\log x)^2 + 1} + C$ (B) $\frac{x}{x^2 + 1} + C$ (C) $\frac{xe^x}{1 + x^2} + C$ (D) $\frac{x}{(\log x)^2 + 1} + C$

Ans. D

Sol. -

Q12. $\int \left(\int e^x \left(\ln x + \frac{2}{x} - \frac{1}{x^2} \right) dx \right) dx =$

(A) $e^x \ln x + C_1 x + C_2$ (B) $e^x \ln x + C_1 x + C_2$ (C) $\frac{\ln x}{x} + C_1 x + C_2$

(D) None of these

Ans. A

Sol. -

Q13. The intergal $\int \left(1 + x - \frac{1}{x} \right) e^{x+\frac{1}{x}} dx$ is equal to -

(A) $(x+1)e^{x+\frac{1}{x}} + c$ (B) $-xe^{x+\frac{1}{x}} + c$ (C) $(x-1)e^{x+\frac{1}{x}} + c$ (D) $xe^{x+\frac{1}{x}} + c$

Ans. D

Sol. $\int \left(1 + x - \frac{1}{x} \right) \cdot e^{x+\frac{1}{x}} dx$

$$= \int e^{x+\frac{1}{x}} \left(1 + x \left(1 - \frac{1}{x^2} \right) \right) dx$$

$$= \int \underbrace{1}_{II} \cdot \underbrace{e^{x+\frac{1}{x}}}_I dx + \int x \cdot e^{x+\frac{1}{x}} \left(1 - \frac{1}{x^2} \right) dx$$

$$= x \cdot e^{x+\frac{1}{x}} - \int x \cdot e^{x+\frac{1}{x}} \left(1 - \frac{1}{x^2} \right) dx + \int xe^{x+\frac{1}{x}} \left(1 - \frac{1}{x^2} \right) dx$$

$$= xe^{x+\frac{1}{x}} + C$$

Q14. $\int \frac{\sin 3x + \sin 4x - \sin 6x}{1 + \cos 2x + \cos 4x + \cos 6x}$ is equal to (where C is indefinite integration constant)

(A) $\frac{1}{3} \ln |\sec 3x| + \frac{1}{2} \ln |\sec 2x| + \ln |\sec x| + C$ (B) $\frac{1}{3} \ln |\sec 3x| - \frac{1}{2} \ln |\sec 2x| - \ln |\sec x| + C$

(C) $C - \frac{1}{3} \ln |\sec 3x| + \frac{1}{2} \ln |\sec 2x| + \ln |\sec x|$

(D) $C - \frac{1}{3} \ln |\sec 3x| - \frac{1}{2} \ln |\sec 2x| - \ln |\sec x|$

Ans. B

Sol. -

Q15. $\int \sec^2(\sec \theta + \tan \theta)^2 d\theta$

- (A) $\frac{(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$
(B) $\frac{(\sec \theta + \tan \theta)}{3} [2 + 4 \tan \theta (\sec \theta + \tan \theta)] + C$
(C) $\frac{(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$
(D) $\frac{3(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$

Ans. C

Sol. -

Q16. $I = \int \frac{(\sin 2x)^{1/3} d(\tan^{1/3} x)}{\sin^{2/3} x + \cos^{2/3} x} =$

- (A) $\frac{1}{2^{2/3}} \ln(1 + \tan^{1/3} x) + C$ (B) $\ln(1 + \tan^{2/3} x) + C$ (C) $21/3 \ln(1 + \tan^{2/3} x) + C$
(D) $\frac{1}{2^{2/3}} \ln(1 + \tan^{2/3} x) + C$

Ans. D

Sol.
$$\begin{aligned} & \int \frac{(\sin 2x)^{\frac{1}{3}}}{\sin^{\frac{2}{3}} x + \cos^{\frac{2}{3}} x} d\left(\tan^{\frac{1}{3}} x\right) \\ &= 2^{\frac{1}{3}} \int \frac{\sin^{\frac{1}{3}} x \cos^{\frac{1}{3}} x}{\left(\tan^{\frac{2}{3}} x + 1\right) \cos^{\frac{2}{3}} x} d\left(\tan^{\frac{1}{3}} x\right) \\ &= \frac{2^{\frac{1}{3}}}{2} \int \frac{2 \tan^{\frac{1}{3}} x}{\left(1 + \left(\tan^{\frac{1}{3}} x\right)^2\right)} d\left(\tan^{\frac{1}{3}} x\right) \\ &= \frac{1}{2^{\frac{2}{3}}} \ln\left(1 + \tan^{\frac{2}{3}} x\right) + C \end{aligned}$$

Q17. If the integral $\int \frac{5 \tan x}{\tan x - 2} dx = x + a$ then a is equal to :

- (A) 2 (B) -1 (C) -2 (D) 1

Ans. A

Sol. -

Q18. $\frac{1 + x^7}{x(1 + x^7)} dx =$

- (A) $\ln x + \frac{2}{7} \ln(1 + x^7) + c$ (B) $\ln x - \frac{2}{7} \ln(1 - x^7) + c$ (C) $\ln x - \frac{2}{7} \ln(1 + x^7) + c$

(D) $\ln x + \frac{2}{7} \ln(1 - x^7) + c$

Ans. C

Sol. -

Q19. Primitive of $\frac{3x^4 - 1}{(x^4 + x + 1)^2}$ w.r.t. x is :

(A) $\frac{x}{x^4 + x + 1} + c$ (B) $-\frac{x}{x^4 + x + 1} + c$ (C) $\frac{x + 1}{x^4 + x + 1} + c$ (D) $-\frac{x + 1}{x^4 + x + 1} + c$

Ans. B

Sol. -

Q20. The evaluation of $\int \frac{px^{f+2q-1} - qx^{q-1}}{x^{2p+2q} + 2x^{(p+q)} + 1} dx$ is

(A) $-\frac{x^p}{x^{p+q} + 1} + C$ (B) $\frac{x^q}{x^{p+q} + 1} + C$ (C) $-\frac{x^q}{x^{p+q} + 1} + C$ (D) $\frac{x^p}{x^{p+q} + 1} + C$

Ans. C

Sol. -

Q21. $\int \frac{dx}{\sqrt[3]{x^{5/2}(x+1)^{7/2}}}$

(A) $-\left(\frac{x+1}{x}\right)^{1/6} + C$ (B) $6\left(\frac{x+1}{x}\right)^{-1/6} + C$ (C) $\left(\frac{x}{x+1}\right)^{5/6} + C$
 (D) $-\left(\frac{x}{x+1}\right)^{5/6} + C$

Ans. B

Sol. -

Q22. The integral $\int \frac{2x^{12} + 5x^9}{(x^5 + x^3 + 1)^3} dx$ is equal to :
 (where c is an arbitrary constant)

(A) $\frac{x^5}{2(x^5 + x^3 + 1)^2} + C$ (B) $\frac{-x^{10}}{2(x^5 + x^3 + 1)^2} + C$ (C) $\frac{-x^5}{2(x^5 + x^3 + 1)^2} + C$
 (D) $\frac{x^{10}}{2(x^5 + x^3 + 1)^2} + C$

Ans. D

Sol. $\int \frac{(2x^{12} + 5x^9)dx}{x^{15}(1 + x^{-2} + x^{-5})^3}$
 $1 + x^{-2} + x^{-5} = t$

$$I = - \int \frac{dt}{t^3}$$

$$= \frac{1}{2t^2} + C = \frac{x^{10}}{2(x^5 + x^3 + 1)^2} + C$$

Q23. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx$ is equal to

(A) (B) (C) (D)

(Where C is a constant of integration)

(A) $\frac{1}{1 + \cot^3 x} + C$ (B) $\frac{-1}{1 + \cot^3 x} + C$ (C) $\frac{1}{3(1 + \tan^3 x)} + C$ (D) $\frac{-1}{3(1 + \tan^3 x)} + C$

Ans. D

Sol. -

Q24. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2}$ equals

(A) $\frac{1}{n(n+1)}(1+nx^n)^{1-\frac{1}{n}} + K$ (B) $\frac{1}{(n+1)}(1+nx^n)^{1-\frac{1}{n}} + K$

(C) $\frac{1}{n(n+1)}(1+nx^n)^{1+\frac{1}{n}} + K$ (D) $\frac{1}{(n+1)}(1+nx^n)^{1+\frac{1}{n}} + K$

Ans. A

Sol. -

Q25. $\int \frac{x^2(1-\ln x)}{\ln^4 x - x^4} dx$ equals

(A) $\frac{1}{2} \ln\left(\frac{x}{\ln x}\right) - \frac{1}{4} \ln(\ln^2 x - x^2) + C$ (B) $\frac{1}{4} \ln\left(\frac{\ln x - x}{\ln x + x}\right) - \frac{1}{2} \tan^{-1}\left(\frac{\ln x}{x}\right) + C$

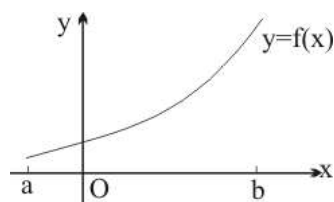
(C) $\frac{1}{4} \ln\left(\frac{\ln x + x}{\ln x - x}\right) + \frac{1}{2} \tan^{-1}\left(\frac{\ln x}{x}\right) + C$ (D) $\frac{1}{4} \left(\ln\left(\frac{\ln x - x}{\ln x + x}\right) + \tan^{-1}\left(\frac{\ln x}{x}\right) \right) + C$

Ans. B

Sol. -

Q26. The graph of the function $y = f(x)$, where $f : [a, b] \rightarrow \mathbb{R}$ and α is a negative constant and b a positive constant, is shown.

The function $F : [a, b] \rightarrow \mathbb{R}$ is defined by $F(t) = \int_0^t f(x) dx$. Then $F(t) > 0$ for



- (A) $t \in [a, b] \setminus \{0\}$ (B) $t \in [a, 0)$ only (C) $t \in [0, b]$ only (D) $t \in (0, b]$ only

Ans. D

Sol. -

Q27. If $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} = 0$, where C_0, C_1, C_2 are all real, the equation $C_2x^2 + C_1x + C_0 = 0$ has :

- (A) atleast one root in $(0, 1)$ (B) one root in $(1, 2)$ and other in $(3, 4)$
 (C) one root in $(-1, 1)$ & the other in $(-5, -2)$ (D) both roots imaginary

Ans. A

Sol. -

Q28. Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$
 Then, for an arbitrary constant C , the value of $J - I$ equals

- (A) $\frac{1}{2} \ln \left(\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1} \right) + C$ (B) $\frac{1}{2} \ln \left(\frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1} \right) + C$ (C) $\frac{1}{2} \ln \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + C$
 (D) $\frac{1}{2} \ln \left(\frac{e^{4x} + e^{2x} + 1}{e^{4x} - e^{2x} + 1} \right) + C$

Ans. C

Sol. -

Q29. If $g(x) = \int_0^x \cos^4 t \, dt$, then $g(x + \pi)$ equals

- (A) $g(x) + g(\pi)$ (B) $g(x) - g(\pi)$ (C) $g(x)g(\pi)$ (D) $[g(x)/g(\pi)]$

Ans. A

Sol.

$$\begin{aligned} g(x + \pi) &= \int_0^{x+\pi} \cos^4 t \, dt \\ &= \int_0^x \cos^4 t \, dt + \int_x^{x+\pi} \cos^4 t \, dt \\ &= g(x) + \int_0^\pi \cos^4 t \, dt = g(x) + g(\pi) \end{aligned}$$

Q30. If $g(x) = \int_1^x e^{t^2}$ then the value of $\int_3^{x^3} e^{t^2}$ equals

- (A) $g(x^3) - g(3)$ (B) $g(x^3) + g(3)$ (C) $g(x^3) - 3$ (D) $g(x^3) - 3g(x)$

Ans. A

Sol. -

Q31. Given $\int_0^{\pi/2} \frac{dx}{1 + \sin x + \cos x} = \ln 2$, then the value of the def. integral. $\int_0^{\pi/2} \frac{\sin x}{1 + \sin x + \cos x} dx$ is equal to

- (A) $\frac{1}{2} \ln 2$ (B) $\frac{\pi}{2} - \ln 2$ (C) $\frac{\pi}{2} - \frac{1}{2} \ln 2$ (D) $\frac{\pi}{2} + \ln 2$

Ans. C

Sol. -

Q32. Let $F(x) = \left(\frac{e^{\sin x}}{x} \right)$, $x > 0$. If $\int_1^4 \frac{3}{x} e^{\sin x^3} dx = F(k) - F(1)$, then one of the possible values of k, is-

- (A) 64 (B) 15 (C) 16 (D) 63

Ans. A

Sol. -

Q33. Let $F(x) = f(x) + f\left(\frac{1}{x}\right)$, where $f(x) = \int_1^x \frac{\log t}{1+t} dt$. Then $F(e)$ equals :

- (A) $1/2$ (B) 0 (C) 1 (D) 2

Ans. A

Sol. -

Q34. The value of $\int_0^1 \left(\prod_{r=1}^n (x+r) \right) \left(\sum_{k=1}^n \frac{1}{x+k} \right) dx$ equals

- (A) n (B) n! (C) (n+1)! (D) n · n!

Ans. D

Sol. The given integrand is perfect differential coeff. of $\prod_{r=1}^n (x+r)$

$$\Rightarrow \left[\prod_{r=1}^n (x+r) \right]_0^1 = (n+1)! - n! = n \cdot n!$$

Alternatively: $(x+1)(x+2)(x+3).....(x+n) = t$

so that when $x = 1$, $t = (n+1)!$

if $x = 0$ then $t = n!$

$$[\ln(x+1) + \ln(x+2) + \dots + \ln(x+n)] = \ln t$$

$$\therefore \left(\frac{1}{(x+1)} + \frac{1}{(x+2)} + \dots + \frac{1}{(x+n)} \right) dx = \frac{1}{t} dt \Rightarrow I = \int_{n!}^{(n+1)!} \frac{1}{t} t dt$$

$$\text{Hence } I = \int_{\ln n!}^{(n+1)!} e^t dt = e^t \Big|_{\ln n!}^{\ln(n+1)!} = e^{\ln(n+1)!} - e^{\ln n!} = (n+1)! - n! = n \cdot n!$$

Q35.

The value(s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ is(are)

(A) $\frac{22}{7} - \pi$ (B) $\frac{2}{105}$ (C) 0 (D) $\frac{71}{15} - \frac{3\pi}{2}$

Ans. A

Sol. Let's use the classic substitution for a $1+x^2$ denominator: $x = \tan \theta \Rightarrow dx = \sec^2 \theta d\theta$

$$\frac{1}{1+x^2} dx = \frac{1}{1+\tan^2 \theta} \cdot \sec^2 \theta d\theta = d\theta$$

$$I = \int_0^{\pi/4} \tan^4 \theta (1 - \tan \theta)^4 d\theta$$

$$I = \int_0^{\pi/4} (\tan^8 \theta - 4 \tan^7 \theta + 6 \tan^6 \theta - 4 \tan^5 \theta + \tan^4 \theta) d\theta$$

Instead of standard reduction formulas, we can use a beautiful identity inherent to tangents:
 $\tan^n \theta + \tan^{n-2} \theta = \tan^{n-2} \theta (\tan^2 \theta + 1) = \tan^{n-2} \theta \sec^2 \theta$

1. **Pair the 8th and 6th powers:** $(\tan^8 \theta + \tan^6 \theta) = \tan^6 \theta \sec^2 \theta$ *(leaves $5 \tan^6 \theta$ remaining)
2. **Pair the 7th and 5th powers:** $-4(\tan^7 \theta + \tan^5 \theta) = -4 \tan^5 \theta \sec^2 \theta$
3. **Pair the remaining 6th power:** $5(\tan^6 \theta + \tan^4 \theta) = 5 \tan^4 \theta \sec^2 \theta$ *(we added $5 \tan^4 \theta$, so we must subtract $4 \tan^4 \theta$ to balance the original $+1 \tan^4 \theta$)*
4. **Pair the new 4th power:** $-4(\tan^4 \theta + \tan^2 \theta) = -4 \tan^2 \theta \sec^2 \theta$ *(we introduced $-4 \tan^2 \theta$, so add $+4 \tan^2 \theta$)*
5. **Pair the last terms:** $4(\tan^2 \theta + 1) - 4 = 4 \sec^2 \theta - 4$

$$I = \int_0^{\pi/4} [(\tan^6 \theta - 4 \tan^5 \theta + 5 \tan^4 \theta - 4 \tan^2 \theta + 4) \sec^2 \theta - 4] d\theta$$

$$I = \left[\frac{\tan^7 \theta}{7} - \frac{4 \tan^6 \theta}{6} + \frac{5 \tan^5 \theta}{5} - \frac{4 \tan^3 \theta}{3} + 4 \tan \theta \right]_0^{\pi/4} - \int_0^{\pi/4} 4 d\theta$$

$$I = \left(\frac{1}{7} - \frac{2}{3} + 1 - \frac{4}{3} + 4 \right) - [4\theta]_0^{\pi/4}$$

$$I = \frac{22}{7} - 4 \left(\frac{\pi}{4} \right)$$

$$I = \frac{22}{7} - \pi$$

Q36. $\int_{-1/2}^{1/2} \left[\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right]^{1/2} dx$ has the value equal to

(A) 4 (B) $4\ln\frac{3}{4}$ (C) $3\ln\frac{3}{4}$ (D) $4\ln\frac{4}{3}$

Ans. D

Sol. -

Q37. Let $[.]$ denote the greatest integer function then the value of $\int_0^{1.5} x[x^2] dx$ is :

(A) $\frac{5}{4}$ (B) 0 (C) $\frac{3}{2}$ (D) $\frac{3}{4}$

Ans. D

Sol. -

Q38. The value of $\int_1^a [x] f(x) dx$, $a > 1$, where $[x]$ denotes the greatest integer not exceeding x is –

(A) $[a] f(a) - \{f(1) + f(2) + \dots + f([a])\}$ (B) $[a] f([a]) - \{f(1) + f(2) + \dots + f(a)\}$
 (C) $a f([a]) - \{f(1) + f(2) + \dots + f(a)\}$ (D) $a f(a) - \{f(1) + f(2) + \dots + f([a])\}$

Ans. A

Sol. -

Q39. The integral $\int_0^\pi \sqrt{1 + 4\sin^2 \frac{x}{2} - 4\sin \frac{x}{2}} dx$ equals:

(A) $4\sqrt{3} - 4$ (B) $4\sqrt{3} - 4 - \frac{\pi}{3}$ (C) $\pi - 4$ (D) $\frac{2\pi}{3} - 4 - 4\sqrt{3}$

Ans. B

Sol. $I = \int_0^\pi \sqrt{1 + 4\sin^2 \frac{x}{2} - 4\sin \frac{x}{2}} dx$

$$\Rightarrow I = \int_0^\pi \sqrt{\left(2\sin \frac{x}{2} - 1\right)^2} dx$$

$$\Rightarrow I = \int_0^\pi \left|2\sin \frac{x}{2} - 1\right| dx$$

$$\Rightarrow I = \int_0^{\pi/3} \left(1 - 2\sin \frac{x}{2}\right) dx + \int_{\pi/3}^\pi \left(2\sin \frac{x}{2} - 1\right) dx$$

$$\begin{aligned}
&= x + 4 \cos \frac{x}{2} \Big|_0^\pi + \left[- \left(4 \cos \frac{x}{2} + x \right) \right]_{\pi/3}^\pi \\
&= \left[\left(\frac{\pi}{3} + 4 \frac{\sqrt{3}}{2} \right) - (0 + 4) \right] - \left[(0 + \pi) - \left(4 \frac{\sqrt{3}}{2} + \frac{\pi}{3} \right) \right] \\
&= 4\sqrt{3} - 4 - \frac{\pi}{3}
\end{aligned}$$

Q40. The integral $\int_{-\frac{1}{2}}^{\frac{1}{2}} \left([x] + \ln \left(\frac{1+x}{1-x} \right) \right) dx$ equals, where $[]$ denotes greatest integer function

(A) $-\frac{1}{2}$ (B) 0 (C) 1 (D) $2\ln\left(\frac{1}{2}\right)$

Ans. A

Sol. -

Q41. $\int_{-1}^1 \frac{x^3 + |x| + 1}{x^2 + 2|X| + 1} dx = a \ln 2 + b$ then-

(A) $a = 2 ; b = 1$ (B) $a = 2 ; b = 0$ (C) $a = 3 ; b = -2$ (D) $a = 4 ; b = -1$

Ans. B

Sol. -

Q42. The value of $\int_{-2}^3 |1 - x^2| dx$ is-

(A) $28/3$ (B) $14/3$ (C) $7/3$ (D) $1/3$

Ans. A

Sol. -

Q43. $\int_{-3\pi/2}^{-\pi/2} [(x + \pi)^3 + \cos^2(x + 3\pi)] dx$ is equal to -

(A) $(\pi^4/32) + (\pi/2)$ (B) $\pi/2$ (C) $(\pi/4) - 1$ (D) $\pi^4/32$

Ans. B

Sol. -

Q44. If $I = \int_0^{\pi/2} \ln(\sin x) dx$ then $\int_{-\pi/4}^{\pi/4} \ln(\sin x + \cos x) dx =$

- (A) $\frac{I}{2}$ (B) $\frac{I}{4}$ (C) $\frac{I}{\sqrt{2}}$ (D) I

Ans. A

Sol. -

Q45. The value of the definite integral $\int_0^{\pi/2} \sin x \sin 2x \sin 3x \, dx$ is equal to :

- (A) $\frac{1}{3}$ (B) $-\frac{2}{3}$ (C) $-\frac{1}{3}$ (D) $\frac{1}{6}$

Ans. D

Sol. -

Q46. The value of $\frac{2\pi}{\pi} [2 \cos x] dx$ where $[\cdot]$ represents the greatest integer function is

- (A) $-\frac{5\pi}{6}$ (B) $-\frac{\pi}{2}$ (C) $-\pi$ (D) none

Ans. B

Sol. -

Q47. If $f(a+b-x) = f(x)$, then $\int_a^b x f(x) dx$ is equal to-

- (A) $\frac{a+b}{2} \int_b^a f(a+b-x) dx$ (B) $\frac{a+b}{2} \int_a^b f(b-x) dx$ (C) $\frac{a+b}{2} \int_a^b f(x) dx$
 (D) $\frac{b-a}{2} \int_a^b f(x) dx$

Ans. C

Sol. -

Q48. If $f(y) = e^y$, $g(y) = y$; $y > 0$ and $F(t) = \int_0^t f(t-y) dy$, then

- (A) $F(t) = t e^{-t}$ (B) $F(t) = 1 - e^{-1} (1+t)$ (C) $F(t) = e^t - (1+t)$ (D) $F(t) = t e^t$

Ans. C

Sol. $f(y) = e^y$, $g(y) = y$; $y > 0$

and $F(t) = \int_0^t f(t-y) g(y) dy$

$$= \int_0^t e^{t-y} y dy = e^t \int_0^t e^{-y} y dy = e^t [-ye^{-y} - e^{-y}]_0^t$$

$$= -e^t [te^{-t} + e^{-t} - 0 - 1] = e^t - (1 + t)$$

Q49. If $f(x) = \frac{e^x}{1+e^x}$, $I_1 = \int_{f(-a)}^{f(a)} x g\{x(1-x)\} dx$ and $I_2 = \int_{f(-a)}^{f(a)} g\{x(1-x)\} dx$, then the value of $\frac{I_2}{I_1}$ is-

- (A) 2 (B) -3 (C) -1 (D) 1

Ans. A

Sol. -

Q50. The value of $\int_{-\pi}^{\pi} \frac{\cos^2}{1+a^x} dx, a > 0$, is -

- (A) $a\pi$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{a}$ (D) $2\frac{\pi}{2}$

Ans. B

Sol. -

Q51. The value of the integral, $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x} + \sqrt{x}} dx$ is -

- (A) $\frac{3}{2}$ (B) 2 (C) 1 (D) $\frac{1}{2}$

Ans. A

Sol. -

Q52. The value of $I = \int_0^{\pi/2} \frac{(\sin x + \cos x)^2}{\sqrt{1 + \sin 2x}} dx$ is-

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. -

Q53. If $\int_0^{\pi} x f(\sin x) dx = A \int_0^{\pi/2} f(\sin x) dx$, then A is-

- (A) 0 (B) π (C) $\pi/4$ (D) 2π

Ans. B

Sol. -

Q54. $\int_{-\pi}^{\pi} \frac{2x(1 + \sin x)}{1 + \cos^2 x} dx$

- (A) π^2 (B) $\pi^2/4$ (C) $\pi/8$ (D) $\pi^2/8$

Ans. A

Sol. -

Q55. The integral $\int_2^4 \frac{\log x^2}{\log x^2 + \log(36 - 12x + x^2)} dx$ is equal to :

- (A) 2 (B) 4 (C) 1 (D) 6

Ans. C

Sol. -

Q56. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + 2^x} dx$ is

- (A) 4π (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. -

Q57. The value of $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)} dx$ is :

- (A) $\frac{1}{4} \ln \frac{3}{2}$ (B) $\frac{1}{2} \ln \frac{3}{2}$ (C) $\ln \frac{3}{2}$ (D) $\frac{1}{6} \ln \frac{3}{2}$

Ans. A

Sol. -

Q58. Let $T > 0$ be a fixed real number. Suppose f is a continuous function such that for all $x \in \mathbb{R}$

$f(x + T) = f(x)$. If $I = \int_0^T f(x) dx$ then the value of $\int_3^{3+3T} f(2x) dx$ is

- (A) $\frac{3}{2}I$ (B) $2I$ (C) $3I$ (D) $6I$

Ans. C

Sol. -

Q59. The value of the definite integral $\int_{19}^{37} (\{x\}^2 + 3(\sin 2\pi x)) dx$ where $\{x\}$ denotes the fractional part function.

- (A) 0 (B) 6 (C) 9 (D) 18

Ans. C

Sol. -

Q60. $\int_{\pi}^{10\pi} |\sin x| dx =$

- (A) 9 (B) 10 (C) 18 (D) 20

Ans. C

Sol. -

Q61. Let $f(x) = \int_1^x \sqrt{2-t^2} dt$. Then the real roots of the equation $x^2 - f'(x) = 0$ are

- (A) ± 1 (B) $\pm \frac{1}{\sqrt{2}}$ (C) $\pm \frac{1}{2}$ (D) 0 and 1

Ans. A

Sol. -

Q62. Variable x and y are related by equation $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$. The value of $\frac{d^2y}{dx^2}$ is equal to

- (A) $\frac{y}{\sqrt{1+y^2}}$ (B) y (C) $\frac{2y}{\sqrt{1+y^2}}$ (D) $4y$

Ans. B

Sol. -

Q63. Let $f(x) = \int_{-1}^x e^{t^2} dt$ and $h(x) = f(1+g(x))$, where $g(x)$ is defined for all x , $g'(x)$ exists for all x , and $g(x) < 0$ for $x > 0$. If $h'(1) = e$ and $g'(1) = 1$, then the possible values which $g(1)$ can take

- (A) 0 (B) -1 (C) -2 (D) -4

Ans. C

Sol. -

Q64.

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 0$, then $\lim_{n \rightarrow \infty} n^2 \int_0^{\frac{1}{n}} f(t) dt$ equals

- (A) $\frac{1}{2}f'(0)$ (B) $f'(0)$ (C) $2f'(0)$ (D) 0

Ans. A

Sol. -

Q65.

If a, b and c are real numbers then the value of $\lim_{t \rightarrow 0} \ln \left(\frac{1}{t} \int_0^t (1 + a \sin bx)^{c/x} dx \right)$ equals

- (A) abc (B) $\frac{ab}{c}$ (C) $\frac{bc}{a}$ (D) $\frac{ca}{b}$

Ans. A

Sol. -

Q66.

$\lim_{x \rightarrow 0} \frac{\int_0^x \sin t^2 dt}{x(1 - \cos x)}$ equals

- (A) $\frac{1}{3}$ (B) 3 (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

Ans. D

Sol. -

Q67.

The value of $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t dt}{x \sin x}$ is-

- (A) 0 (B) 3 (C) 2 (D) 1

Ans. D

Sol. -

Q68. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function having $f(2) = 6$, $f'(2) = \left(\frac{1}{48}\right)$. Then $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$ equals -

$\int_6^{f(x)} \frac{4t}{x-2} dt$ equals -

- (A) 24 (B) 36 (C) 12 (D) 18

Ans. D

Sol. -

Q69. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t) dt}{x^2 - \frac{\pi^2}{16}}$ equals

- (A) $\frac{8}{\pi} f(2)$ (B) $\frac{2}{\pi} f(2)$ (C) $\frac{2}{\pi} f\left(\frac{1}{2}\right)$ (D) $4 f(2)$

Ans. A

Sol. -

Q70. The value of $\lim_{x \rightarrow 0} \int_0^x \frac{t \ln(1+t)}{t^4 + 4} dt$ is

- (A) 0 (B) $\frac{1}{12}$ (C) $\frac{1}{24}$ (D) $\frac{1}{64}$

Ans. B

Sol. -

Q71. If $f'(x) = f(x) + \int_0^1 f(x) dx$ and given $f(0) = 1$, then $\int f(x) dx$ is equal to :

- (A) $\frac{2}{3-e} e^x + \left(\frac{3-e}{1-e}\right)x + C$ (B) $\frac{2}{3-e} e^x + \left(\frac{1-e}{3-e}\right)x + C$
 (C) $\frac{3}{2-e} e^x + \left(\frac{1-e}{3-e}\right)x + C$ (D) $\frac{2}{2-e} e^x + \left(\frac{1-e}{3+e}\right)x + C$

Ans. B

Sol. -

Q72. The set of values of 'a' which satisfy the equation $\int_0^2 (t - \log_2 a) dt = \log_2 \left(\frac{4}{a^2}\right)$ is

- (A) $a \in \mathbb{R}$ (B) $a \in \mathbb{R}^+$ (C) $a < 2$ (D) $a > 2$

Ans. B

Sol. -

Q73. If $\int_{\sin x}^1 t^2(f(t)) dt = (1 - \sin x)$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is

- (A) $1/3$ (B) $1/\sqrt{3}$ (C) 3 (D) $\sqrt{3}$

Ans. C

Sol. -

Q74. Let $p(x)$ be a function defined on R such that $p'(x) = p'(1-x)$, for all $x \in [0, 1]$, $p(0) = 1$ and $p(1) = 41$. $\int_0^1 p(x) dx$ Then equals :

- (A) $\sqrt{41}$ (B) 21 (C) 41 (D) 42

Ans. B

Sol. -

Q75. Let f be a non-negative function defined on the interval $[0, 1]$. If

$$\int_0^x \sqrt{1 - (f'(t))^2} dt \int_0^x f(t) dt, 0 \leq x \leq 1 \text{ and } f(0) = 0, \text{ then}$$

- (A) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$ (B) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
 (C) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$ (D) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

Ans. C

Sol. -

Q76. Let f be a real-valued function defined on the interval $(-1, 1)$ such that $e^{-x} f(x) = 2 +$

$$\int_0^x \sqrt{t^4 + 1} dt, \text{ for all}$$

$x \in (-1, 1)$, and let f^{-1} be the inverse function of f .

Then $(f^{-1})'(2)$ is equal to

- (A) 1 (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{e}$

Ans. B

Sol. -

Q77. $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}\sqrt{4n}}$ is equal to

- (A) 2 (B) 4 (C) $2(\sqrt{2} - 1)$ (D) $2\sqrt{2} - 1$

Ans. A

Sol. -

Q78. $f(x) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2 x^2}, x > 0$ is equal to

- (A) $x \tan^{-1}(x)$ (B) $\tan^{-1}(x)$ (C) $\frac{\tan^{-1}(x)}{x}$ (D) $\frac{\tan^{-1}(x)}{x^2}$

Ans. C

Sol. -

Q79. $\lim_{n \rightarrow \infty} \frac{1^P + 2^P + 3^P + \dots + n^P}{n^{P+1}} =$

- (A) 1 (B) $\frac{1}{P+1}$ (C) $\frac{1}{P+2}$ (D) p^2

Ans. B

Sol. -

Q80. $\lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sec^2 \frac{1}{n^2} + \frac{2}{n^2} \sec^2 \frac{4}{n^2} + \dots + \frac{1}{n} \sec^2 1 \right]$ equals -

- (A) $\frac{1}{2} \sec 1$ (B) $\frac{1}{2} \cos ec 1$ (C) $\tan 1$ (D) $\frac{1}{2} \tan 1$

Ans. D

Sol. -

Q81. $\lim_{n \rightarrow \infty} \left(\frac{(n+1)(n+2) \dots 3n}{n^{2n}} \right)^{1/n}$ is equal to :

- (A) $\frac{9}{e^2}$ (B) $3 \log 3 - 2$ (C) $\frac{18}{e^4}$ (D) $\frac{27}{e^2}$

Ans. D

Sol. -

Q82. If $I_1 = \int_0^1 2^{x^2} dx$, $I_2 = \int_0^1 2^{x^3} dx$, $I_3 = \int_1^2 2^{x^2} dx$ and $I_4 = \int_1^2 2^{x^3} dx$ then -

- (A) $I_2 > I_1$ (B) $I_1 > I_2$ (C) $I_3 = I_4$ (D) $I_3 > I_4$

Ans. B

Sol. -

Q83. Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$. Then which one of the following is true ?

- (A) $1 > \frac{2}{3}$ and $J > 2$ (B) $1 < \frac{2}{3}$ and $J < 2$ (C) $1 < \frac{2}{3}$ and $J > 2$
(D) $1 > \frac{2}{3}$ and $J < 2$

Ans. B

Sol. -

Q84. If $I_n = \int (\sin x)^n dx$, $n \in \mathbb{N}$ then $5I_4 - 6I_6$ is equal to

- (A) $\sin x \cdot (\cos x)^5 + C$ (B) $\sin 2x \cdot \cos 2x + C$ (C) $\frac{\sin 2x}{8} [\cos^2 2x + 1 - 2 \cos 2x] + C$
 (D) $\frac{\sin 2x}{8} [\cos^2 2x + 1 + 2 \cos 2x] + C$

Ans. C

Sol. -

Q85. Let $I_n = \int \tan^n x \, dx$, ($n > 1$). If $I_4 + I_6 = a \tan^5 x + bx^5 + C$, where C is a constant of integration, then the ordered pair (a, b) is equal to

- (A) $\left(\frac{1}{5}, -1\right)$ (B) $\left(-\frac{1}{5}, 0\right)$ (C) $\left(-\frac{1}{5}, 1\right)$ (D) $\left(\frac{1}{5}, 0\right)$

Ans. D

Sol. -

Exercise J02

Q1. $\int \frac{\sin 2x - \sin 2k}{\sin x - \sin k + \cos x - \cos k} dx$

Ans. $(\sin x - \cos x) + (\sin k + \cos k)x + C$

Sol. -

Q2. $\int \frac{x^2 + 3}{x^6(x^2 + 1)} dx$

Ans. $C - \frac{2}{x} + \frac{2}{3} \frac{1}{x^3} - \frac{2}{5} \frac{2}{x^5} - 2 \tan^{-1} x$

Sol. -

Q3. $\int \sin x \cos x \cos 2x \cos 4x dx$

Ans. $-\frac{1}{64} \cos 8x + C$

Sol. -

Q4. $\int \frac{\ln \left(\ln \left(\frac{1+x}{1-x} \right) \right)}{1-x^2} dx$

Ans. $\frac{1}{2} \left[\ln \left(\frac{1+x}{1-x} \right) \cdot \ln \left(\ln \frac{1+x}{1-x} \right) - \ln \left(\frac{1+x}{1-x} \right) \right] + C$

Sol. -

Q5. $\int \left[\left(\frac{x}{e} \right)^2 + \left(\frac{e}{x} \right)^x \right] \ln x dx$

Ans. $\left(\frac{x}{e} \right)^x - \left(\frac{e}{x} \right)^x + C$

Sol. -

Q6. $\int \frac{\log_6 \sqrt[6]{((\sin x) 6^{\cos x})} \cos x}{\sin x} dx$

Ans. $\frac{1}{6} \left(\frac{\ln^2(\sin x)}{\ln 36} + \ln \left(\tan \frac{x}{2} \right) + \cos x \right) + C$

Sol. -

Q7. $\int \frac{x^2 + x}{(e^x + x + 1)} dx$

Ans. $C - \ln(1 + (x+1)e^{-x}) - \frac{1}{1 + (x+1)e^{-1}}$

Sol. -

Q8. Let
$$= \int \frac{f'(x)g(x) - g'(x)f(x)}{(f(x) + g(x))\sqrt{f(x)g(x) - g^2(x)}} dx = \sqrt{m} \tan^{-1} \left(\sqrt{\frac{f(x) - g(x)}{ng(x)}} \right) + C,$$
 where $m, n \in \mathbb{N}$ and 'C' is constant of integration ($g(x) > 0$). Find the value of $(m^2 + n^2)$.

Ans. 8

Sol. -

Q9.
$$\int \frac{\sin^p x}{\cos^{p+2} x} dx$$

Ans.
$$\frac{\tan^{p+1} x}{p+1} + c$$

Sol. -

Q10.
$$\int x^x \ln(ex) dx$$

Ans. $x^x + C$

Sol. -

Q11.
$$\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$$

Ans. $(a+x) \arctan \sqrt{\frac{x}{a}} - \sqrt{ax} + C$

Sol. -

Q12.
$$\int \sin^{-1} \left(\frac{2x+2}{\sqrt{4x^2+8x+13}} \right) dx.$$

Ans. $(x+1) \tan^{-1} \frac{2(x+1)}{3} - \frac{3}{4} \ln(4x^2+8x+13) + C$

Sol. -

Q13.
$$\int \frac{x \ln x}{(x^2-1)} dx$$

Ans. $\arcsin x - \frac{\ln x}{\sqrt{x^2-1}} + C$

Sol. -

Q14.
$$\int x^3 e^{x^2} dx$$

Ans. $\frac{1}{2}(x^2-1)e^{x^2} + c$

Sol. -

Q15.
$$\int \frac{x^5 + 3x^4 - x^3 + 8x^2 - x + 8}{x^2 + 1} dx$$

Ans. $\frac{x^4}{4} + x^3 - x^2 + 5x + \frac{1}{2}\ln(x^2 + 1) + 3\tan^{-1}x + C$

Sol. -

Q16. $\int \frac{(\sqrt{x} + 1)dx}{\sqrt{x}(\sqrt[3]{x} + 1)}$

Ans. $6\left[\frac{t^4}{4} - \frac{t^2}{2} + t + \frac{1}{2}\ln(1 + t^2) - \tan^{-1}t\right] + C$ where $t = x^{1/6}$

Sol. -

Q17. $\int \sqrt{x^2 - 8x + 7}dx$

Ans. $-\frac{9}{2}\log[x - 4\sqrt{x^2 - 8x + 7}] + \frac{1}{2}(x - 4)\sqrt{x^2 - 8x + 7} + c$

Sol. -

Q18. $\int \frac{e^x(2 - x^2)}{(1 - x)\sqrt{1 - x^2}}dx$

Ans. $e^x\sqrt{\frac{1+x}{1-x}} + c$

Sol. -

Q19. $\int \frac{e^{\cos x}(x \sin^3 x + \cos x)}{\sin^2 x}dx$

Ans. $C - e^{\cos x}(x + \operatorname{cosec} x)$

Sol. -

Q20.

(a) $\int \frac{dx}{\cot \frac{x}{2} \cdot \cot \frac{x}{3} \cdot \cot \frac{x}{6}}$; (b) $\int \frac{\tan(\ln x) \tan\left(\ln \frac{x}{2}\right) \tan(\ln 2)}{x} dx$

Ans. (a) $2\ln\left(\sec \frac{x}{2}\right) - 3\ln\left(\sec \frac{x}{3}\right) - 6\ln\left(\sec \frac{x}{6}\right) + C$; (b) $\ln\left(\frac{\sec(\ln x)}{\sec(\ln(x/2))x^{\tan(9\ln 2)}}\right) + C$

Sol. -

Q21. $\int \sqrt{\frac{\operatorname{cosec} x - \cot x}{\operatorname{cosec} x + \cot x}} \cdot \frac{\sec x}{\sqrt{1 + 2\sec x}} dx$

Ans. $\sin^{-1}\left(\frac{1}{2}\sec^2 \frac{x}{2}\right) + C$

Sol. The first radical is a well-known standard identity:

$$\sqrt{\frac{\operatorname{csc} x - \cot x}{\operatorname{csc} x + \cot x}} = \operatorname{csc} x - \cot x = \tan\left(\frac{x}{2}\right)$$

Now, let's rewrite the second term using : $\frac{\sec x}{\sqrt{1 + 2\sec x}} = \frac{1}{\sqrt{\cos^2 x + 2\cos x}}$

So, the integral simplifies directly to: $I = \int \tan\left(\frac{x}{2}\right) \cdot \frac{1}{\sqrt{\cos^2 x + 2 \cos x}} dx$

Apply the Substitution $t = \tan\left(\frac{x}{2}\right)$

$$I = \int \frac{2t}{\sqrt{3 - 2t^2 - t^4}} dt$$

Let $y = t^2$, so $dy = 2t dt$.

$$I = \int \frac{dy}{\sqrt{3 - 2y - y^2}}$$

$$I = \int \frac{dy}{\sqrt{2^2 - (y + 1)^2}}$$

$$I = \sin^{-1}\left(\frac{y + 1}{2}\right) + C$$

$$I = \sin^{-1}\left(\frac{\tan^2\left(\frac{x}{2}\right) + 1}{2}\right) + C$$

Q22. $\int \frac{\tan 2\theta}{\sqrt{\cos^6 \theta + \sin^6 \theta}} d\theta$

Ans. $\ln\left(\frac{1 + \sqrt{1 + 3 \cos^2 2\theta}}{\cos 2\theta}\right) + C$

Sol. -

Q23. $\int \frac{a^2 \sin^2 x + b^2 \cos^2 x}{a^4 \sin^2 x + b^4 \cos^2 x} dx$

Ans. $\frac{1}{a^2 + b^2} \left(x + \tan^{-1} \left(\frac{a^2 \tan x}{b^2} \right) \right) + C$

Sol. -

Q24. $\int \frac{\cos^2 x}{1 + \tan x} dx$

Ans. $\frac{1}{4} \ln(\cos x + \sin x) + \frac{x}{2} + \frac{1}{8}(\sin 2x + \cos 2x) + C$

Sol. -

Q25. $\int (\sin x)^{-11/3} (\cos x)^{-1/3} dx$

Ans. $\frac{3(1 + 4 \tan^2 x)}{8(\tan x)^{8/3}} + C$

Sol. -

Q26. $\int \frac{dx}{\sec x + \cos ecx}$

Ans. $\frac{1}{2} \left[\sin x - \cos x - \frac{1}{\sqrt{2}} \ln \tan \left(\frac{x}{2} + \frac{\pi}{8} \right) \right] + C$

Sol. -

Q27. $\int \frac{dx}{\sin x + \sec x}$

Ans. $\frac{1}{2\sqrt{3}} \ln \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} + \arctan(\sin x + \cos x) + C$

Sol. -

Q28. $\int \frac{\cos x - \sin x}{7 - 9\sin 2x} dx$

Ans. $\frac{1}{24} \ln \frac{(4 + 3\sin x + 3\cos x)}{4 - 3\sin x - 3\cos x} + C$

Sol. -

Q29. $\int \frac{\sqrt{\cot x} - \sqrt{\tan x}}{1 + 3\sin 2x} dx$

Ans. $\tan^{-1} \left(\frac{\sqrt{2} \sin 2x}{\sin x + \cos x} \right) + C$

Sol. -

Q30. $\int \frac{3x^2 + 1}{(x^2 - 1)^3} dx$

Ans. $C - \frac{x}{(x^2 - 1)^2}$

Sol. -

Q31. Let $f(x)$ is a quadratic function such that $f(0) = 1$ and $\int \frac{f(x)dx}{x^2(x+1)^3}$ is a rational function,
find the value of $f'(0)$.

Ans. 3

Sol. -

Q32. $\int \frac{4x^5 - 7x^4 + 8x^3 - 2x^2 + 4x - 7}{x^2(x^2 + 1)^2} dx$

Ans. $4\ln x + \frac{7}{x} + 6\tan^{-1}(x) + \frac{6x}{1+x^2} + C$

Sol. -

Q33. $\int \frac{(ax^2 - b)dx}{x\sqrt{c^2x^2 - (ax^2 - b)^2}}$

Ans. $\sin^{-1}\left(\frac{ax^2}{cx} + k\right)$

Sol. -

Q34. $\int \frac{dx}{\left(x + \sqrt{x(1+x)}\right)^2}$

Ans. $2\ln\frac{t}{2t+1} + \frac{1}{2t+1} + C$ when $t = x + \sqrt{x^2 + x}$

Sol. -

Q35. $\int \frac{5x^4 + 4x^5}{(x^5 + x + 1)^2} dx$

Ans. $C - \frac{x+1}{x^5+x+1}$ or $C \frac{x^5}{x^5+x+1}$

Sol. -

Q36. For any natural number m, evaluate

$$\int (x^{3m} + x^{2x} + x^m)(2x^{2m} + 3x^m + 6)^{\frac{1}{m}}, \text{ where } x > 0$$

Ans. $\frac{1}{6(m+1)}(2x^{3m} + 3x^{2m} + 6x^m)^{\frac{m+1}{m}} + C$

Sol. -

Q37. $\int \frac{dx}{(x-\alpha)\sqrt{(x-\alpha)(x-\beta)}}$

Ans. $\frac{-2}{\alpha-\beta} \cdot \sqrt{\frac{x-\beta}{x-\alpha}} + C$

Sol. $\frac{1}{(x-\alpha)\sqrt{(x-\alpha)(x-\beta)}} = \frac{1}{(x-\alpha)^2\sqrt{\frac{(x-\alpha)(x-\beta)}{(x-\alpha)^2}}} = \frac{1}{(x-\alpha)^2} \cdot \frac{1}{\sqrt{\frac{x-\beta}{x-\alpha}}}$

Now, the integral looks like this: $I = \int \frac{1}{\sqrt{\frac{x-\beta}{x-\alpha}}} \cdot \frac{dx}{(x-\alpha)^2}$

$$u = \frac{x-\beta}{x-\alpha}$$

$$du = -\frac{\alpha-\beta}{(x-\alpha)^2} dx \implies \frac{dx}{(x-\alpha)^2} = \frac{du}{\beta-\alpha}$$

$$I = \int \frac{1}{\sqrt{u}} \left(\frac{du}{\beta-\alpha} \right) = \frac{1}{\beta-\alpha} \int u^{-1/2} du$$

$$I = \frac{1}{\beta-\alpha} \cdot (2u^{1/2}) + C = \frac{2}{\beta-\alpha} \sqrt{u} + C$$

$$I = \frac{2}{\beta - \alpha} \sqrt{\frac{x - \beta}{x - \alpha}} + C \dots$$

Q38. $\int \frac{x^3 + x + 1}{x^4 + x + 1} dx$

Ans. $\frac{1}{2} \ln(x^2 + x + 1) - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x + 1}{\sqrt{3}} \right) + \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x - 1}{\sqrt{3}} \right) - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x^2 + 1}{\sqrt{3}} \right) + C$

Sol. -

Q39. $\int \frac{(1 + x^2) dx}{1 - 2x^2 \cos \alpha + x^4} \alpha \in (0, \pi)$

Ans. $\frac{1}{2} \left(\cos ec \frac{\alpha}{2} \right) \cdot \tan^{-1} \left(\left(\frac{x^2 - 1}{2x} \right) \cos ec \frac{\alpha}{2} \right)$

Sol. -

Q40. Evaluate: $\int_0^1 e^{\ln \tan^{-1} x} \cdot \sin^{-1}(\cos x) dx$.

Ans. $\frac{\pi^2}{8} - \frac{\pi}{4}(1 - \ln 2) + \frac{1}{2}$

Sol. -

Q41. Prove that: **(a)** $\int_{\alpha}^{\beta} \sqrt{x - \alpha(\beta - x)} dx = \frac{(\beta - \alpha)^2 \pi}{8}$ **(b)**

$$\int_{\alpha}^{\beta} \sqrt{\frac{x - \alpha}{\beta - x}} dx = (\beta - \alpha) \frac{\pi}{2}$$

(c) $\int_{\alpha}^{\beta} \frac{dx}{x \sqrt{(x - \alpha)(\beta - x)}} = \frac{\pi}{\sqrt{\alpha \beta}}$ where $\alpha, \beta > 0$ **(d)**

$$\int_{\alpha}^{\beta} \frac{x \cdot dx}{\sqrt{(x - \alpha)(\beta - x)}} = \alpha + \beta \frac{\pi}{2} = \text{where } \alpha < \beta$$

Ans. **(a)** $\frac{1}{2} \left[e^{\pi/2} (\cos 1 + \sin 1) - 1 \right]$; **(b)** $e^{1+e} + e^{1-e} + e^{-e} - e^e + e - e^{-1}$

Sol. -

Q42. Evaluate: **(a)** $\int_0^{\pi/5} e^x \left\{ \cos(\sin x) \cos^2 \frac{x}{2} + \sin(\sin x) \sin^2 \frac{x}{2} \right\} dx$ **(b)**

$$\int_1^e \left\{ (1 + x)e^x + (1 - x)e^{-x} \right\} \cdot \ln x dx$$

Ans. -

Sol. -

Q43. If $P = \int_0^{\infty} \frac{x^2}{1+x^4} dx$; $Q = \int_0^{\infty} \frac{x dx}{1+x^4}$ and $R = \int_0^{\infty} \frac{dx}{1+x^4}$ then prove that

$$(a) Q = \frac{\pi}{4} \quad (b) P = R, \quad (c) P - \sqrt{2}Q + R = \frac{\pi}{2\sqrt{2}}$$

Ans. -

Sol.
$$\int_0^{\infty} \frac{x^2}{1+x^4} dx$$

$$\text{Let } x^2 = \tan \theta \Rightarrow 2x dx = \sec^2 \theta d\theta$$

$$\Rightarrow P = \int_0^{\pi/2} \frac{\tan \theta}{1 + \tan^2 \theta} \cdot \frac{\sec^2 \theta d\theta}{2\sqrt{\tan \theta}}$$

$$P = \frac{1}{2} \int_0^{\pi/2} \sqrt{\tan \theta} d\theta$$

Apply king prop.

$$P = \frac{1}{2} \int_0^{\pi/2} \sqrt{\cot \theta} d\theta$$

$$\Rightarrow 2P = \frac{1}{2} \int_0^{\pi/2} (\sqrt{\tan \theta} + \sqrt{\cot \theta}) d\theta$$

$$\Rightarrow 2P = \frac{1}{4} \int_0^{\pi/2} \frac{(1 + \tan \theta)}{\sqrt{\tan \theta}} d\theta$$

$$\Rightarrow P = \frac{\pi}{4}$$

$$R = \int_0^{\infty} \frac{dx}{1+x^4}$$

$$\text{Let } x^2 = \tan \theta$$

$$2x dx = \sec^2 \theta d\theta$$

$$R = \int_0^{\pi/2} \frac{1}{\sec^2 \theta} \frac{\sec^2 \theta}{2\sqrt{\tan \theta}} d\theta$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi/2} \sqrt{\cot \theta} d\theta$$

Apply king prop.

$$\Rightarrow R = \frac{1}{2} \int_0^{\pi/2} \sqrt{\tan \theta} d\theta$$

$$\Rightarrow R = \frac{\pi}{4}$$

$$\Rightarrow P = R$$

$$\begin{aligned} P - \sqrt{2}Q + R &= \int_0^{\infty} \frac{(x^2 - \sqrt{2}x + 1)}{x^4 + 1} dx \\ &= \int_0^{\infty} \frac{(x^2 - \sqrt{2}x + 1) dx}{(x^2 - \sqrt{2}x + 1)(x^2 + \sqrt{2}x + 1)} = \int_0^{\infty} \frac{dx}{x^2 + \sqrt{2}x + 1} \\ &= \int_0^{\infty} \frac{dx}{\left(x + \frac{1}{\sqrt{2}}\right)^2 + \frac{1}{2}} = \sqrt{2} \left[\tan^{-1} \left(\frac{x + \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} \right) \right]_0^{\infty} = \frac{\pi}{2\sqrt{2}} \end{aligned}$$

Q44. For $a \geq 2$, if the value of the definite integral $\int_0^{\infty} \frac{dx}{a^2 + (x - (1/x))^2} = \frac{\pi}{5050}$. Find the value of a .

Ans. 2525

Sol.
$$\int_0^{\infty} \frac{dx}{a^2 + (x - (1/x))^2} = \frac{\pi}{5050}$$

Let $x = \frac{1}{t}$

$$= \int_{\infty}^0 \frac{-1}{t^2} \cdot \frac{1}{\left(a^2 + \left(\frac{1}{t} - t\right)^2\right)} dt$$

$$I = \int_0^{\infty} \frac{1}{x^2 \left(a^2 + \left(x - \frac{x}{x} \right)^2 \right)} dx$$

$$\Rightarrow 2I = \int_0^{\infty} \frac{\left(1 + \frac{1}{x^2} \right)}{a^2 + \left(x - \frac{1}{x} \right)^2} \text{ Let } x - \frac{1}{x} = \theta$$

$$2I = \int_0^{\infty} \frac{dx}{a^2 + \theta^2} = \frac{1}{a} \left[\tan^{-1} \left(x - \frac{1}{x} \right) \right]_0^{\infty}$$

$$\Rightarrow I = \frac{1}{2a} \left[\frac{\pi}{2} + \frac{\pi}{2} \right] = \frac{\pi}{2a}$$

$$\Rightarrow a = 2525$$

Q45. If $\int_0^{\pi} \sqrt{(\cos x + \cos 2x + \cos 3x)^2 + (\sin x + \sin 2x + \sin 3x)^2} dx$ has the value equal to $\left(\frac{\pi}{k} + \sqrt{w} \right)$ where k and w are positive integers find the value of $(k^2 + w^2)$.

Ans. 153

Sol. -

Q46. $\int_0^{\frac{\ln 3}{2}} \frac{e^x + 1}{e^{2x+1}} dx$

Ans. $\frac{1}{2} \left[\frac{\pi}{6} + \ln 3 - \ln 2 \right]$

Sol. -

Q47. Let $I = \int_0^1 \frac{2 + 3x + 4x^2}{2\sqrt{1+x+x^2}} dx$. Find the value of I^2 .

Ans. 3

Sol. -

Q48. Show that the sum of the two integrals $\int_{-4}^{-5} e^{(x+5)^2} dx + 3 \int_{1/3}^{2/3} e^{9(x-2/3)^2} dx$ is zero.

Ans. -

Sol. -

Q49. $\int_0^{\pi/2} \tan^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right] dx$

Ans. $\frac{3\pi^2}{16}$

Sol. -

Q50.
$$\int_{\frac{\sqrt{3a^2+b^2}}{2}}^{\frac{\sqrt{a^2+b^2}}{2}} \frac{x \cdot dx}{\sqrt{(x^2 - a^2)(b^2 - x^2)}}$$

Ans. $\frac{\pi}{12}$

Sol. -

Q51. (a) Show that $\int_0^\infty \frac{dx}{x^2 + 2x \cos \theta + 1} = 2 \int_0^1 \frac{dx}{x^2 + 2x \cos \theta + 1}$

(b) Evaluate: $f(\theta) = \int_0^\infty \frac{\tan^{-1} x}{x^2 + 2x \cos \theta + 1} dx, \theta \in (0, \pi)$

Ans. (b) $\frac{\pi\theta}{4 \sin \theta}$

Sol. -

Q52. Let $I = \int_0^{\pi/2} \frac{\cos x + 4}{3 \sin x + 4 \cos x + 25} dx$ and $J = \int_0^{\pi/2} \frac{\sin x + 3}{3 \sin x + 4 \cos x + 25} dx$.

If $25I = a\pi + b \ln \frac{c}{d}$ where a, b, c and $d \in \mathbb{N}$ and $\frac{c}{d}$ is not a perfect square of a rational then find the value of $(a + b + c + d)$.

Ans. 62

Sol. -

Q53. (a) Let $\beta(n) = \int_0^{n\pi} \sqrt{1 - \sin t} dt$. Find the value of $\beta(2) - \beta(1)$.

Ans. (a) 4

Sol. -

Q54.
$$\int_0^{\pi/2} \sqrt{\frac{1 - \sin 2x}{1 + \sin 2x}} dx$$

Ans. $\ln 2$

Sol. -

Q55. Find the value of the definite integral $\int_0^\pi \left| \sqrt{2} \sin x + 2 \cos x \right| dx$.

Ans. $2\sqrt{6}$

Sol. $\sqrt{2} \sin x + 2 \cos x = \sqrt{6} \sin(x + \alpha)$; Where $\alpha = \tan^{-1} \left(\frac{2}{\sqrt{2}} \right) = \tan^{-1}(\sqrt{2})$. Since both terms are positive, α lies in the first quadrant $\left(0 < \alpha < \frac{\pi}{2} \right)$.

Substitute this back into the integral: $I = \int_0^{\pi} |\sqrt{6} \sin(x + \alpha)| dx = \sqrt{6} \int_0^{\pi} |\sin(x + \alpha)| dx$

Let $u = x + \alpha$, which means $du = dx$.

$$I = \sqrt{6} \int_{\alpha}^{\pi+\alpha} |\sin u| du$$

$$I = \sqrt{6} \int_0^{\pi} |\sin u| du$$

$$I = \sqrt{6} \int_0^{\pi} \sin u du$$

$$I = \sqrt{6} \times 2 = 2\sqrt{6}$$

Q56.
$$\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7 + 3x^6 - 10x^5 - 7x^3 - 12x^2 + x + 1}{x^2 + 2} dx$$

Ans.
$$\frac{\pi}{2\sqrt{2}} - \frac{16\sqrt{2}}{5}$$

Sol. -

Q57.
$$\int_{-2}^2 \frac{x^2 - x}{\sqrt{x^2 + 4}} dx$$

Ans.
$$4\sqrt{2} - 4\ln(\sqrt{2} + 1)$$

Sol. -

Q58. Evaluate :
$$\int_{-1}^1 \frac{(2x^{332} + x^{998} + 4x^{1668}) \sin x^{691}}{1 + x^{666}} dx$$

Ans.
$$\frac{\pi + 4}{666}$$

Sol. -

Q59. Evaluate
$$\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2 - \cos \left(|x| + \frac{\pi}{3} \right)} dx.$$

Ans.
$$\frac{4\pi}{\sqrt{3}} \tan^{-1} \left(\frac{1}{2} \right)$$

Sol. -

Q60. Let $u = \int_0^{\pi/4} \left(\frac{\cos x}{\sin x + \cos x} \right)^2 dx$ and $v = \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{\cos x} \right)^2 dx$. Find the value of $\frac{f}{u}$.

Ans. 4

Sol. -

Q61. $\int_0^{\pi/4} \frac{x dx}{\cos x (\cos x + \sin x)}$

Ans. $\frac{\pi}{8} \ln 2$

Sol. -

Q62. $\int_0^1 \frac{\sin^{-1} \sqrt{x}}{x^2 - x + 1} dx$

Ans. $\frac{\pi^2}{6\sqrt{3}}$

Sol. -

Q63. $\int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin \left(\frac{\pi}{4} + x \right)} dx$

Ans. $\frac{\pi^2}{6\sqrt{3}}$

Sol. -

Q64. $\int_0^{\pi} \frac{(ax + b) \sec x \tan x}{4 + \tan^2 x} dx$ (a, b > 0)

Ans. $\frac{(a\pi + 2b)\pi}{3\sqrt{3}}$

Sol. -

Q65. $\int_0^{\pi} \frac{(2x + 3) \sin x}{(1 + \cos^2 x)} dx$

Ans. $\frac{\pi(\pi + 3)}{2}$

Sol. -

Q66. Let $I_n = \int_{-n}^n \left(\{x+1\} \cdot \{x^2+2\} + \{x^2+2\} \{x^3+4\} \right) dx$ where $\{ \}$ denotes the fractional part of x . Find I_1 .

Ans. $2/3$

Sol. -

Q67. $\int_0^{\pi} \frac{x^2 \sin 2x \cdot \sin \left(\frac{\pi}{2} \cdot \cos x \right)}{2x - \pi} dx$

Ans. $\frac{8}{\pi}$

Sol. -

Q68. $\int_0^{2\pi} \frac{dx}{2 + \sin 2x}$

Ans. $\frac{2\pi}{\sqrt{3}}$

Sol. -

Q69. If f is an even function then prove that $\int_0^{\frac{\pi}{2}} f(\cos 2x) \cos x \, dx = \sqrt{2} \int_0^{\pi/4} f(\sin 2x) \cos x \, dx$

Ans. -

Sol. -

Q70. Show that $\int_0^{p+q\pi} |\cos x| \, dx = 2q + \sin p$ where $q \in \mathbb{N}$ & $-\frac{\pi}{2} < p < \frac{\pi}{2}$

Ans. -

Sol. -

Q71. (a) For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-10, 10]$ by

$$f(x) = \begin{cases} x - [x] & \text{if } [x] \text{ is odd} \\ 1 + [x] - x & \text{if } [x] \text{ is even} \end{cases}$$

Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \, dx$ is

(b) Let f be a real-valued function defined on the interval $(0, \infty)$ by $f(x) = \ln x + \int_0^x \sqrt{1 + \sin t} \, dt$. Then which of

the following statement(s) is (are) true?

(A) $f''(x)$ exists for all $x \in (0, \infty)$

(B) $f'(x)$ exists for all $x \in (0, \infty)$ and f' is continuous on $(0, \infty)$, but not differentiable on $(0, \infty)$

(C) there exists $\alpha > 1$ such that $|f'(x)| < |f(x)|$ for all $x \in (\alpha, \infty)$

(D) there exists $\beta > 1$ such that $|f(x)| + |f'(x)| \leq \beta$ for all $x \in (0, \infty)$

Ans. (a) 4, (b) BC

Sol. -

Q72.
$$\lim_{n \rightarrow \infty} n^2 \int_{-1/n}^{1/n} (2010 \sin x + 2012 \cos x) |x| dx.$$

Ans. 2012

Sol. -

Q73. 72 If $y(x) = \int_{\pi^2/16}^{x^2} \frac{\cos x \cdot \cos \sqrt{\theta}}{1 + \sin^2 \sqrt{\theta}} d\theta$ then find $\frac{dy}{dx}$ at $x = \pi$.

Ans. 2π

Sol. -

Q74. A continuous real function f satisfies $f(2x) = 3f(x) \forall x \in \mathbb{R}$

If $\int_0^1 f(x) dx = 1$, then compute the value of definite integral $\int_0^2 f(x) dx$.

Ans. 5

Sol. -

Q75. Let $f(x)$ be a function defined on $\mathbb{R}$ such that $f'(x) = f'(3-x) \forall x \in [0, 3]$

with $f(0) = -32$ and $f(3) = 46$. Then find the value of $\int_0^3 f(x) dx$.

Ans. 21

Sol. -

Q76. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function which satisfies $f(x) = \int_0^x f(t) dt$. Then the value of $f(\ln 5)$ is

Ans. 0

Sol. -

Q77. Evaluate:
$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=0}^{n-1} \left[k \int_k^{k+1} \sqrt{(x-k)(k+1-x)} dx \right]$$

Ans. $\frac{\pi}{16}$

Sol. -

Q78. Evaluate: (a) $\left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \left(1 + \frac{3^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{1/n}$; (b)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left[\frac{1}{n+1} + \frac{2}{n+2} + \dots + \frac{3n}{4n} \right]$$

Ans. (a) $2e^{\frac{\pi-4}{2}}$; (b) $3 - \ln 4$

Sol. -

Q79. Consider a function $f(n) = \frac{1}{1+n^2}$. Let $\alpha_n = \frac{1}{n} \sum_{r=1}^n r \left(\frac{r}{n} \right)$ and $\beta_n = \frac{1}{n} \frac{1}{n} \sum_{r=0}^{n-1} f\left(\frac{r}{n}\right)$ for $n = 1, 2, 3, \dots$

Also $\alpha = \lim_{n \rightarrow \infty} \alpha_n$ & $\beta = \lim_{n \rightarrow \infty} \beta_n$. Then prove (a) $\alpha_n < \beta_n$ (b) $\alpha = \beta$ (c) $\alpha_n < \frac{\pi}{4} < \beta_n$

Ans. -

Sol. -

Q80. Prove the inequalities:

(a) $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$

(b) $2e^{-1/4} < \int_0^2 e^{x^2-x} dx < 2e^2$

(c) $\frac{1}{3} < \int_0^1 x^{(\sin x + \cos x)^2} dx < \frac{1}{2}$

(d) $\frac{1}{2} \leq \int_0^2 \frac{dx}{2+x^2} \leq \frac{5}{6}$

Ans. -

Sol. -

Q81. If a_1, a_2 and a_3 are the three values of α which satisfy the equation

$$\int_0^{\pi/2} (\sin x + a \cos x)^3 dx - \frac{4a}{\pi-2} \int_0^{\pi/2} x \cos x dx = 2$$

then find the value of $1000(a_1^2 + a_2^2 + a_3^2)$.

Ans. 5250

Sol. -

Q82. Determine a positive integer $n \leq 5$, such that $\int_0^1 e^x (x-1)^n dx = 16 - 6e$.

Ans. $n = 3$

Sol. -

Q83. Let $U_{10} = \int_0^{\frac{\pi}{2}} x \sin^{10} x$, then find the value of $\left(\frac{100U_{10} - 1}{U_8} \right)$

Ans. 90.

Sol. -

Q84. Find the value of $\frac{5050 \int_0^1 (1 - x^{50})^{100} dx}{\int_0^1 (1 - x^{50})^{101} dx}$

Ans. 5051

Sol. -

Paragraph

Let the function f satisfies $f(x) \cdot f'(-x) = f(-x) \cdot f'(x)$ for all x and $f(0) = 3$.

\n

Q85. The value of $f(x) \cdot f(-x)$ for all x , is

(A) 4 (B) 9 (C) 12 (D) 16

Ans. B

Sol. -

Paragraph

Let the function f satisfies $f(x) \cdot f'(-x) = f(-x) \cdot f'(x)$ for all x and $f(0) = 3$.

\n

Q86. $\int_{-51}^{51} \frac{dx}{3 + f(x)}$ has the value equal to

(A) 17 (B) 34 (C) 102 (D) 0

Ans. A

Sol. -

Paragraph

Let the function f satisfies $f(x) \cdot f'(-x) = f(-x) \cdot f'(x)$ for all x and $f(0) = 3$.

\n

Q87. Number of roots of $f(x) = 0$ in $[-2, 2]$ is

- (A) 0 (B) 1 (C) 2 (D) 4

Ans. A

Sol. -

Paragraph

Consider the integral $I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot 9x}{1 + e^{2 \sin^3 4x}} dx$

\n

Q88. If $I = k \int_0^{\pi/2} \cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x dx$ then 'k' is equal to

- (A) 1 (B) 5 (C) 10 (D) 20

Ans. C

Sol. -

Paragraph

Consider the integral $I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot 9x}{1 + e^{2 \sin^3 4x}} dx$

\n

Q89. If $I = C \cdot \int_0^{\pi/4} \cos 6x \cdot \cos 8x \cdot \cos 2x dx$ then 'c' equals

- (A) 5 (B) 10 (C) 20 (D) $\frac{5}{2}$

Ans. B

Sol. -

Paragraph

Consider the integral $I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot 9x}{1 + e^{2 \sin^3 4x}} dx$

\n

Q90. The value of 'I' equals

- (A) $\frac{5\pi}{4}$ (B) $\frac{5\pi}{16}$ (C) $\frac{5\pi}{32}$ (D) $\frac{5\pi}{8}$

Ans. D

Sol. -

Paragraph

Let $y = f(x)$ be a twice differentiable, non-negative function defined on $[a, b]$. The area $\int_a^b f(x) dx$, $b > a$ bounded by $y = f(x)$, the x-axis and the ordinates at $x = a$ and $x = b$ can be approximated as

$$\int_a^b f(x) dx \cong \frac{(b-a)}{2} \{f(a) + f(b)\}.$$

Since $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, $c \in (a, b)$, a better approximation to $\int_a^b f(x) dx$ can be written as $\int_a^b f(x) dx \cong \frac{(c-a)}{2} \{f(a) + f(c)\} + \frac{(b-c)}{2} \{f(c) + f(b)\} \equiv F(c)$.

If $c = \frac{a+b}{2}$, then this gives : $\int_a^b f(x) dx \cong \frac{b-a}{4} \{f(a) + 2f(c) + f(b)\}$, (1)

\n

Q91.

The approximate value of $\int_0^{\pi/2}$ using rule (1) given above is

- (A) $\frac{\pi}{8\sqrt{2}}(1 + \sqrt{2})$ (B) $\frac{\pi}{4\sqrt{2}}(1 + \sqrt{2})$ (C) $\frac{\pi}{8}(1 + \sqrt{2})$ (D) $\frac{\pi}{4}(1 + \sqrt{2})$

Ans. C

Sol. -

Paragraph

Let $y = f(x)$ be a twice differentiable, non-negative function defined on $[a, b]$. The area $\int_a^b f(x) dx$, $b > a$ bounded by $y = f(x)$, the x-axis and the ordinates at $x = a$ and $x = b$ can be approximated as

$$\int_a^b f(x) dx \cong \frac{(b-a)}{2} \{f(a) + f(b)\}.$$

Since $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, $c \in (a, b)$, a better approximation to $\int_a^b f(x) dx$ can be

written as $\int_a^b f(x)dx \cong \frac{(c-a)}{2} \{f(a) + f(c)\} + \frac{(b-c)}{2} \{f(c) + f(b)\} \equiv F(c)$.

If $c = \frac{a+b}{2}$, then this gives : $\int_a^b f(x) dx \cong \frac{b-a}{4} \{f(a) + 2f(c) + f(b)\}$, (1)

\n

Q92.

If $\lim_{t \rightarrow a} \left\{ \frac{\int_a^t f(x)dx - \frac{(t-a)}{2} (f(t) + f(a))}{(t-a)^3} \right\} = 0$, for each fixed a,

then f (x) is a polynomial of degree utmost

(A) 4 (B) 3 (C) 2 (D) 1

Ans. D

Sol. -

Paragraph

Let $y = f(x)$ be a twice differentiable, non-negative function defined on $[a, b]$. The area $\int_a^b f(x) dx$, $b > a$ bounded by $y = f(x)$, the x-axis and the ordinates at $x = a$ and $x = b$ can be approximated as

$\int_a^b f(x) dx \cong \frac{(b-a)}{2} \{f(a) + f(b)\}$.

Since $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, $c \in (a, b)$, a better approximation to $\int_a^b f(x)$ can be

written as $\int_a^b f(x)dx \cong \frac{(c-a)}{2} \{f(a) + f(c)\} + \frac{(b-c)}{2} \{f(c) + f(b)\} \equiv F(c)$.

If $c = \frac{a+b}{2}$, then this gives : $\int_a^b f(x) dx \cong \frac{b-a}{4} \{f(a) + 2f(c) + f(b)\}$, (1)

\n

Q93. If $f''(x) < 0$, $x \in (a, b)$, then at the point C on $y = f(x)$ for which F(c) is a maximum, $f'(c)$ is given by

(A) $f'(c) = \frac{f(b) - f(a)}{b - a}$ (B) $f'(c) = \frac{f(b) - f(a)}{a - b}$ (C) $f'(c) = \frac{2(f(b) - f(a))}{b - a}$

(D) $f'(c) = 0$

Ans. A

Sol. -

Paragraph

Consider the function $f(x)$ and $g(x)$, both defined from $\mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \frac{x^3}{2} + 11 - x \int_0^x g(t)dt \text{ and } g(x) = x - \int_0^1 f(t)dt, \text{ then -}$$

\n

Q94. Minimum value of $f(x)$ is

- (A) 0 (B) 1 (C) $\frac{3}{2}$ (D) Does not exist

Ans. B

Sol. -

Paragraph

Consider the function $f(x)$ and $g(x)$, both defined from $\mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \frac{x^3}{2} + 11 - x \int_0^x g(t)dt \text{ and } g(x) = x - \int_0^1 f(t)dt, \text{ then -}$$

\n

Q95. The number of points of intersection of $f(x)$ and $g(x)$ is/are -

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. -

Paragraph

Consider the function $f(x)$ and $g(x)$, both defined from $\mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \frac{x^3}{2} + 11 - x \int_0^x g(t)dt \text{ and } g(x) = x - \int_0^1 f(t)dt, \text{ then -}$$

\n

Q96. The area bounded by $g(x)$ with co-ordinate axes is (in square units) :

- (A) $\frac{9}{4}$ (B) $\frac{9}{2}$ (C) $\frac{9}{8}$ (D) None of these

Ans. C

Sol. -

Paragraph

Let $f : \mathbb{R} \rightarrow \left[\frac{3}{4}, \infty\right)$ be a surjective quadratic function with line of symmetry $2x - 1 = 0$ and $f(1) = 1$.

\n

Q97. If $g(x) = \frac{f(x) + f(-x)}{2}$ then $\int \frac{dx}{\sqrt{g(e^x) - 2}}$ is equal to : (Where C is constant of integration)

- (A) $\sec^{-1}(e^{-x}) + C$ (B) $\sec^{-1}(e^x) + C$ (C) $\sin^{-1}(e^{-x}) + C$ (D) $\sin^{-1}(e^x) + C$

Ans. B

Sol. -

Paragraph

Let $f: \mathbb{R} \rightarrow \left[\frac{3}{4}, \infty\right)$ be a surjective quadratic function with line of symmetry $2x - 1 = 0$ and $f(1) = 1$.

\n

Q98. $\int \frac{e^x}{f(e^x)}$

- (A) $\cot^{-1}\left(\frac{2e^x - 1}{\sqrt{3}}\right) + C$ (B) $\frac{2}{\sqrt{3}}\cot^{-1}\left(\frac{2e^x + 1}{\sqrt{3}}\right) + C$ (C) $\tan^{-1}\left(\frac{2e^x + 1}{\sqrt{3}}\right) + C$
 (D) $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2e^x - 1}{\sqrt{3}}\right) + C$

Ans. C

Sol. -

Q99. Suppose $J = \int \frac{\sin^2 x + \sin x}{1 + \sin x + \cos x} dx$ and $K = \int \frac{\cos^2 x + \cos x}{1 + \sin x + \cos x} dx$. If C is an arbitrary constant of integration then which of the following is/are correct :

- (A) $J = \frac{1}{2}(x - (\sin x + \cos x)) + c$ (B) $J = K - (\sin x + \cos x) + C$ (C) $J = x - K + C$
 (D) $K = \frac{1}{2}(x - \sin x + \cos x) + c$

Ans. A, B, C

Sol. $J + K = \int \frac{1 + \sin x + \cos x}{1 + \sin x + \cos x} dx$

$$J + K = x + C \quad \dots(1) \Rightarrow \quad (C)$$

$$\begin{aligned} \text{again} \quad J - K &= \int \frac{(\sin^2 x - \cos^2 x) + \sin x - \cos x}{1 + \sin x + \cos x} dx \\ &= \int \frac{(\sin x - \cos x) + \sin x + \cos x + 1}{1 + \sin x + \cos x} dx \end{aligned}$$

$$J - K = -\cos x - \sin x + C \quad \dots(2)$$

$$\text{hence } J = K - (\sin x + \cos x) + C \Rightarrow (B)$$

Also (1) + (2)

$$2J = x - (\cos x + \sin x) + C$$

$$J = \frac{1}{2}[x - \sin x - \cos x] + C$$

and (1) - (2)

$$2K = x + (\sin x + \cos x) + C$$

$$K = \frac{1}{2}[x + \sin x + \cos x] + C$$

from (1) $J = x - K + C \Rightarrow (C)$

Q100. Let $u = \int_0^\infty \frac{dx}{x^4 + 7x^2 + 1}$ & $v = \int_0^\infty \frac{x^2 dx}{x^4 + 7x^2 + 1}$ then :

(A) $v > u$ (B) $6v = \pi$ (C) $3u + 2v = 5\pi/6$ (D) $u + v = \pi/3$

Ans. D, C, B

Sol. -

Q101. Let $S_n = \sum_{k=1}^n \frac{k^2 + n^2}{n^3}$ and $T_n = \sum_{k=0}^n \frac{k^2 + n^2}{n^3}$ for $n = 1, 2, 3, \dots$. Then,

(A) $S_n < \frac{4}{3}$ (B) $T_n > \frac{4}{3}$ (C) $S_n > \frac{4}{3}$ (D) $T_n < \frac{4}{3}$

Ans. D, C

Sol. -

Q102. If $f(\theta) = \lim_{n \rightarrow \infty} \sum_{r=0}^{n\theta} \frac{2r}{\sqrt{2\theta - 2r}(n\theta + 2r)}$ then :

(A) $f(1) = \frac{\pi}{6}$ (B) $f(\theta) = \frac{\theta}{2} \int_0^\theta \frac{dx}{\sqrt{\theta^2 - \left(x - \frac{\theta}{2}\right)^2}}$ (C) $f(\theta)$ is a constant function

(D) $y = f(\theta)$ is invertible

Ans. A, B, D

Sol. -

Q103. Let $J = \int_{-1}^2 \left(\cot^{-1} \frac{1}{x} + \cot^{-1} x \right) dx$, $K = \int_{-2\pi}^{7\pi} \frac{\sin x}{|\sin x|} dx$. Then which of the following alternative(s) is/are correct?

(A) $2J + 3K = 8\pi$ (B) $4J^2 + K^2 = 26\pi^2$ (C) $2J - K = 3\pi$ (D) $\frac{J}{K} = \frac{2}{5}$

Ans. B, A

Sol. -

Q104. Which of the following function(s) is/are even?

(A) $f(x) = \int_0^x \ln(t + \sqrt{1+t^2}) dt$ (B) $f(x) = \int_0^x \frac{(2^t + 1)}{2^t - 1} dt$
 (C) $h(x) = \int_0^x (\sqrt{1+t+t^2} - \sqrt{1-t+t^2}) dt$ (D) $l(x) = \int_0^x \ln\left(\frac{1-t}{1+t}\right) dt$

Ans. D, C, A

Sol. -

Q105. Let $S_n = \sum_{k=1}^n \frac{n}{n^2 + kn + k^2}$ and $T_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2}$, for $n = 1, 2, 3, \dots$. Then,

(A) $S_n < \frac{\pi}{3\sqrt{3}}$ (B) $S_n > \frac{\pi}{3\sqrt{3}}$ (C) $T_n < \frac{\pi}{3\sqrt{3}}$ (D) $T_n > \frac{\pi}{3\sqrt{3}}$

Ans. A, D

Sol. -

Q106. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \sin x} dx$ $n = 0, 1, 2, \dots$, then

(A) $I_n = I_{n+2}$ (B) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$ (C) $\sum_{m=1}^{10} I_{2m} = 0$ (D) $I_n = I_{n+1}$

Ans. A, B, C

Sol. -

Q107. **Column-I**

(A) $\int \frac{x^4 - 1}{x^2 \sqrt{x^4 + x^2 + 1}} dx$
 (B) $\int \frac{x^2 - 1}{x \sqrt{1 + x^4}} dx$
 (C) $\int \frac{1 + x^2}{(1 - x^2) \sqrt{1 + x^4}} dx$
 (D) $\int \frac{1}{(1 - x^4) \sqrt{\sqrt{1 + x^4} - x^2}} dx$

(A) (A) S; (B) P; (C) Q; (D) R

Column-II

(P) $\ln\left(\frac{(x^2 + 1) + \sqrt{x^4 + 1}}{x}\right) + C$
 (Q) $C - \frac{1}{\sqrt{2}} \ln\left(\frac{\sqrt{x^4 + 1} - \sqrt{2x}}{(x^2 - 1)}\right)$
 (R) $C - \tan^{-1}\left(\sqrt{\sqrt{1 + \frac{1}{x^4}} - 1}\right)$
 (S) $\frac{\sqrt{x^4 + x^2 + 1}}{x} + C$

(B) (A) P; (B) S; (C) Q; (D) R

(C) (A) S; (B) P; (C) R; (D) Q (D) (A) R; (B) Q; (C) P; (D) S

Ans. 5f22b953fddefaba3a41bfe2

Sol. -

Q108. Let $I_1 = \int \tan x \tan(ax + b)dx$ and $I_2 = \int \cot x \cot(ax + b)dx$

Column-I

Column-II

(A) value of I_1 for $a = 1$ is

(P) $x - \cot b \ln \frac{\cos(x-b)}{\cos x} + C$

(B) value of I_2 for $a = 1$ is

(Q) $\cot b \ln \frac{\sin x}{\sin(x+b)} - x + C$

(C) value of I_1 for $a = -1$ is

(R) $\cot b \ln \left(\frac{\cos x}{\cos(x+b)} \right) - x + C$

(D) value of I_2 for $a = -1$ is

(S) $x + \cot b \ln \left(\frac{\sin x}{\sin(b-x)} \right) + C$

(A) (A) Q; (B) R; (C) P; (D) S (B) (A) R; (B) Q; (C) P; (D) S

(C) (A) R; (B) Q; (C) S; (D) P (D) (A) R; (B) P; (C) Q; (D) S

Ans. 5f239205fddefaba3a41ccae

Sol. -

Q109. Let $I = \int \frac{e^x}{e^{4x} + 1} dx$ and $J = \int \frac{e^{-x}}{e^{-4x} + 1} dx$

Then for any arbitrary constant C, match the following

Column-I

Column-II

(A) I

(P) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{e^{2x} - 1}{\sqrt{2}e^x} \right) + C$

(B) J + I

(Q) $\frac{1}{2\sqrt{2}} \ln \left(\frac{e^{2x} - \sqrt{2}e^x + 1}{e^{2x} + \sqrt{2}e^x + 1} \right) + C$

(C) J - I

(R) $\frac{1}{2\sqrt{2}}$

$\left(\tan^{-1} \left(\tan^{-1} \left(\frac{e^{2x} - 1}{\sqrt{2}e^x} \right) - \frac{1}{2} \ln \left(\frac{e^{2x} - \sqrt{2}e^x + 1}{e^{2x} + \sqrt{2}e^x + 1} \right) \right) \right) + C$

(S) $\frac{1}{2\sqrt{2}}$

$\left(\tan^{-1} \left(\tan^{-1} \left(\frac{e^{2x} - 1}{\sqrt{2}e^x} \right) + \frac{1}{2} \ln \left(\frac{e^{2x} - \sqrt{2}e^x + 1}{e^{2x} + \sqrt{2}e^x + 1} \right) \right) \right) + C$

(A) (A) R; (B) P; (C) S (B) (A) R; (B) P; (C) Q

(C) (A) S; (B) P; (C) Q (D) (A) R; (B) S; (C) Q

Ans. 5f239627b9745eba2470be1d

Sol. -

Q110. Column-I

$$(A) \int \frac{x^2(x^6 + x^5 - 1)dx}{(2x^6 + 3x^5 + 2)^2}$$

$$(B) \int \frac{(x^5 + x^4 + x^2)dx}{\sqrt{4x^7 + 5x^6 + 10x^4}}$$

$$(C) \int \frac{(2x^{12} + 5x^9)dx}{(x^5 + x^3 + 1)^3}$$

(where C is the constant of integration.)

Column-II

$$(P) \frac{1}{6} \left(\frac{1}{x^{-3}} + \frac{3}{x^{-3}} \right)^{\frac{1}{2}} + C$$

$$(Q) \frac{1}{2} (1 + x^{-2} + x^{-5})^{-2} + C$$

$$(R) \frac{1}{2} (2x^3 + 3x^2 + 2x^{-3})^{-1} + C$$

$$(S) x \left(\frac{x^3}{25} + \frac{x^2}{20} + \frac{1}{10} \right)^{\frac{1}{2}} + C$$

(A) (A) R, (B) S, (C) Q

(B) (A) R, (B) P, (C) Q

(C) (A) R, (B) S, (C) P

(D) (A) P, (B) S, (C) Q

Ans. 5f2399b0fddefaba3a41cd0d

Sol. -

Q111. Match the integrals in Column I with the values in Column II.

Column I

Column II

$$(A) \int_{-1}^1 \frac{dx}{1+x^2}$$

$$(P) \frac{1}{2} \log \left(\frac{2}{3} \right)$$

$$(B) \int_0^1 \frac{dx}{\sqrt{1-x^2}}$$

$$(Q) 2 \log \left(\frac{2}{3} \right)$$

$$(C) \int_2^3 \frac{dx}{1-x^2}$$

$$(R) \frac{\pi}{3}$$

$$(D) \int_1^2 \frac{dx}{x\sqrt{x^2-1}}$$

$$(S) \frac{\pi}{2}$$

(A) (A) R, (B) S, (C) P, (D) R

(B) (A) S, (B) R, (C) P, (D) P

(C) (A) S, (B) S, (C) P, (D) R

(D) (A) R, (B) P, (C) S, (D) R

Ans. A

Sol.

$$(A) \int_{-1}^1 \frac{dx}{1+x^2} = [\tan^{-1}x]_{-1}^1 = \frac{\pi}{2}$$

$$(B) \int_0^1 \frac{dx}{\sqrt{1-x^2}} = [\sin^{-1}x]_0^1 = \frac{\pi}{2}$$

$$(C) \int_2^3 \frac{dx}{1-x^2} = \left[\frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| \right]_2^3$$

$$= \frac{1}{2} \ln \left(\frac{2}{3} \right)$$

$$(D) \int_1^2 \frac{dx}{x\sqrt{x^2-1}} = (\sec^{-1}x)_1^2 = \frac{\pi}{3} - 0 = \frac{\pi}{3}$$